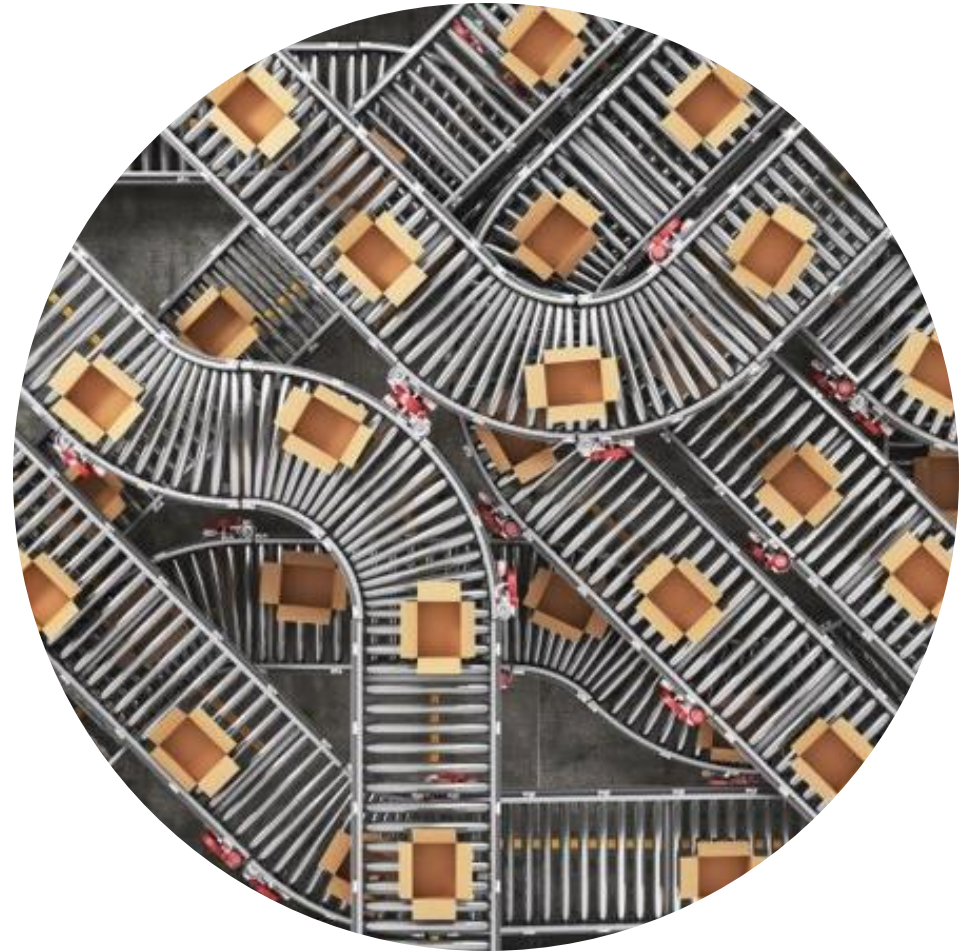


Warehouse PDDL/PDDL+ Model: Single Robot Pick-and-Deliver



Scenario & Planning Problem

Objects

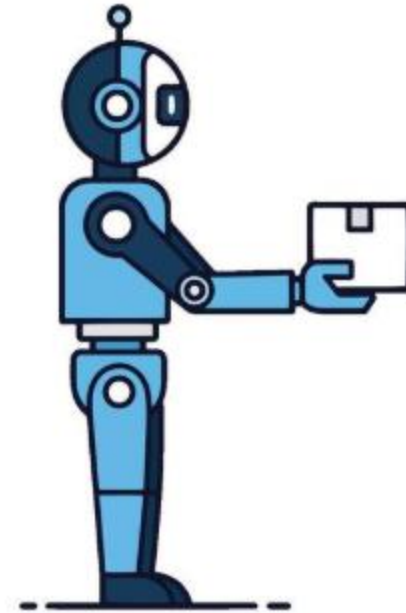
- Robot
- Packages
- Locations

Constraints

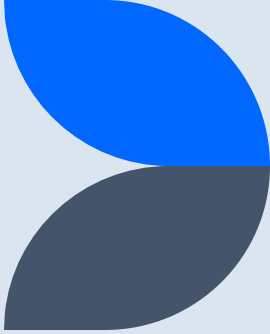
- Single robot
- Single package capacity
- Connected warehouse graph

Tasks

- Implement navigation
- Pick-up package action
- Deliver-package action



PDDL Model Design



Predicates

- At
- Package-at
- Holding
- Free
- Connected
- Goal location
- Delivered

Actions

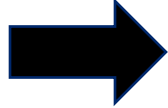
- Move
- Pick up
- Drop off

Requirement	Solution
Explicit robot position	At
Explicit package position	Package-at
Single package capacity	Free + Holding
Explicit transport	Move
Explicit package manipulation	Pick up + Drop off

PDDL+ Extension

Additions

- Modeling time spent during transportation
- Implementing deadline mechanic



New predicates

- Moving
- Transiting
- Overdue

Fluents

- Elapsed-time
- Travel-time
- Move-progress
- Deadline

Process

- Moving-process
- Time-passage

Events

- Deadline-missed

Action split

Move → Start-move
Finish-move

Demo 1: Basic PDDL Plan

Problems defined

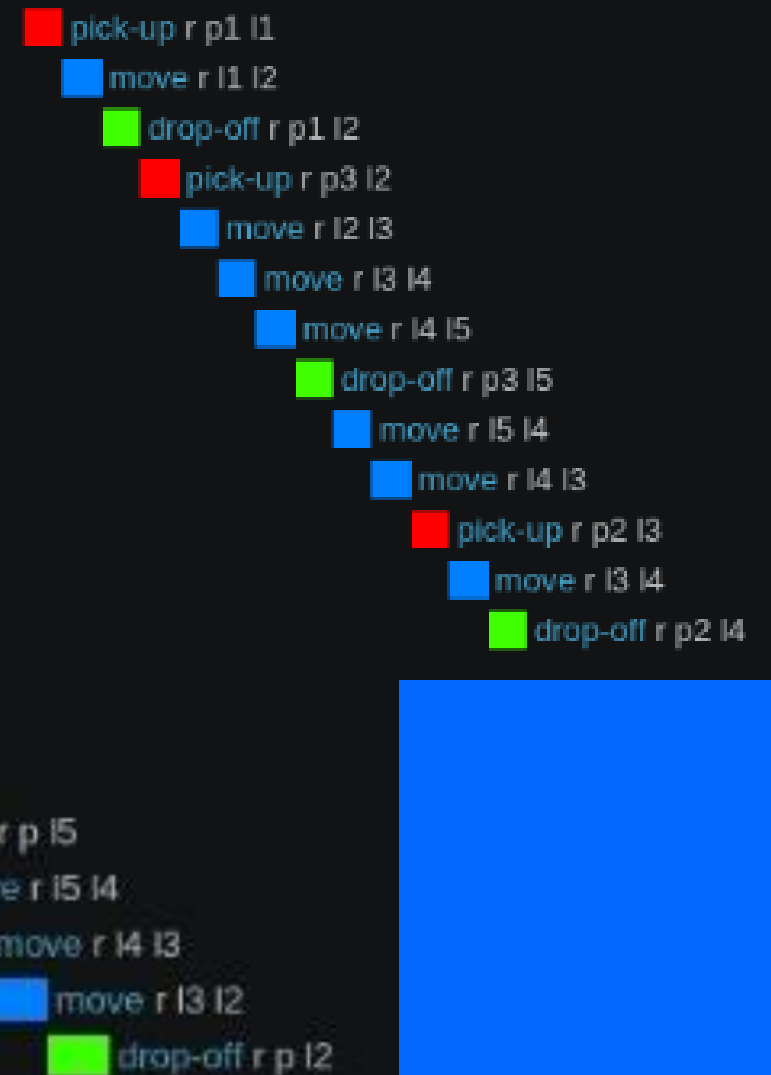
- Single-Package Delivery
- Multi-Package Delivery

Key execution aspects

- Closest Package priority
- Sequential Delivery



Predictable Planner Behavior



Demo 2: PDDL+ Plan with Deadlines

Change of paradigm

- Non-instantaneous actions
- Deadlines

Time taken to execute deliveries becomes **relevant**



Influence on plan **Feasibility**

Discussion

Implementation Difficulties

- Modelling transportation time without durative actions
- Connecting start-move and finish-move safely
- Preventing teleportation

Model Limitations

- Single robot
- Single package capacity
- Sequential deliveries



Future Work & Improvements

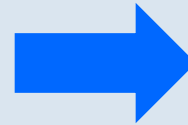
Possible extensions

- Multiple robots
- Multiple package carrying



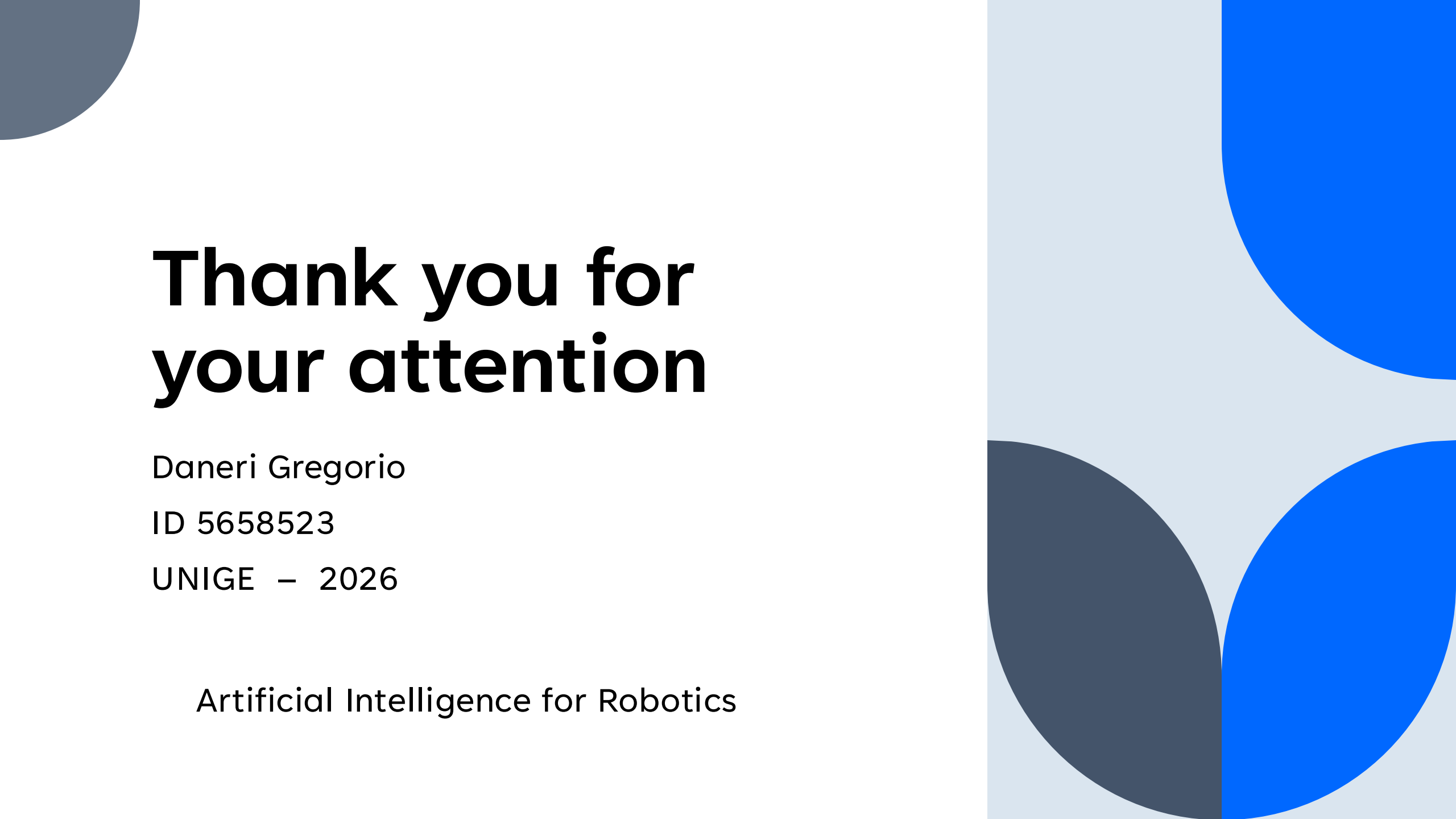
Complications

- Robot collisions
- Package priority
- Maximum payload capacity
- Location precedence



Debatable Solutions & Upgrades

- Payload modelling
- Virtual map
- Fixed paths
- Guarding mechanisms
- Real-time scheduling algorithms
- Route optimization



Thank you for your attention

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Artificial Intelligence for Robotics